

Cosmic topology: is the Universe finite?

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1 Introduction

Question: Is the Universe finite or infinite?

Non-trivial global shape
Closed? Compact? Orientable?
Simply-/Multiply- connected? $\Rightarrow$ *Cosmic topology*

* General relativity concerns only the local geometry of the spacetime manifold, like curvature, and consider the topology as *trivial*, i.e., simply-connected and infinite.

* Quantum processes in the very early Universe may introduce *non-trivial* topology, such as inflation or quantum gravity effects.

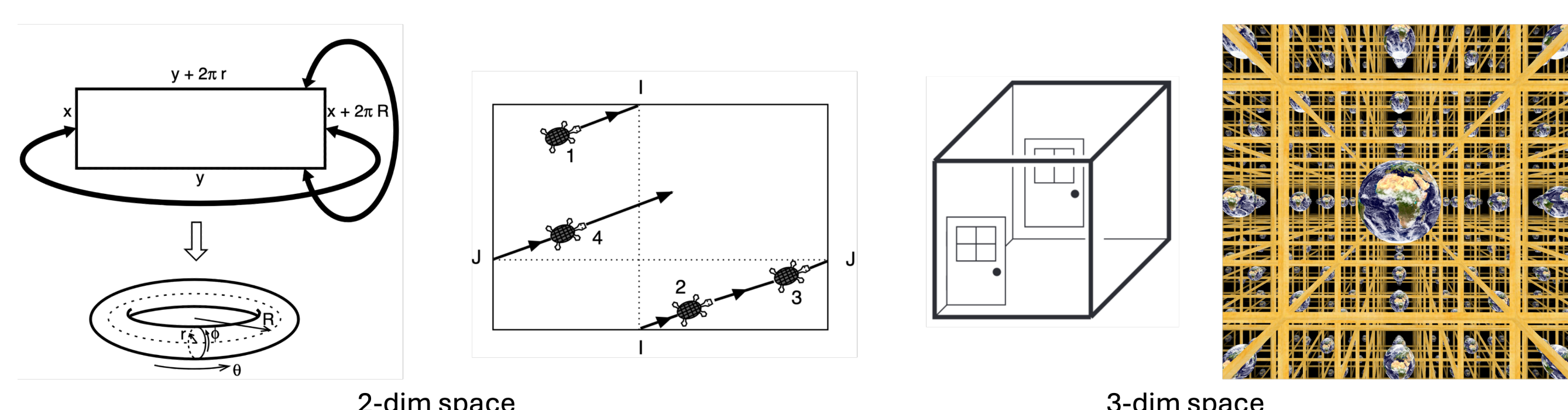
* Brief history[1]:

- ~18th: Infinite, Euclidean space (Kant, Newton)
- 1870s: Finite, unbounded, torus Universe (Riemann, Schwarzschild)
- 1920s: Static, curved, simply-connected space (Einstein, de Sitter)
- 1930s: Expanding, open/closed, multiply-connected (Friedmann, Lemaître)
- 1970s: Probed the observational consequences (Ellis, Luminet)

Torus Universe

Topological properties of space [2]:

- Curvature ($\mathbb{H}^3/\mathbb{R}^3/\mathbb{S}^3$): if open or closed, i.e., unbounded or bounded
- Compact / Non-compact: if finite (can be unbounded) or infinite volume
- Simply-connected / Multiply-connected: if have holes
- Orientable / Non-orientable: if can define left and right consistently
- ★ A standard 3-torus $\mathbb{T}^3$ Universe is the simplest compact flat three-manifold, and it is unbounded, orientable, multiply-connected space.



Observational Signatures

Consider a 3-torus Universe with side-length L , which is the size of the fundamental domain (i.e. basic shape).

- Cosmic Microwave Background (CMB) ”circles-in-the-sky” (2-dim): searching for matching circular patterns on the Last Scattering Surface (LSS, i.e. observable Universe boundary), noting only effective when $L < \text{LSS radius}$.

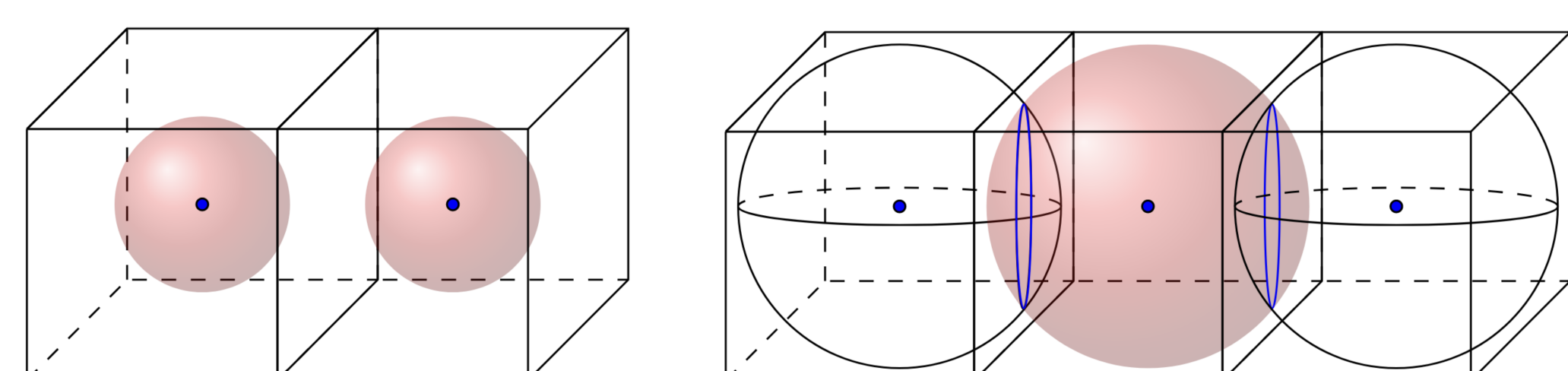


Fig: The LSS compared to the size of a 3-torus Universe. For right the LSS intersects itself and will have circles of matching temperature distributions.

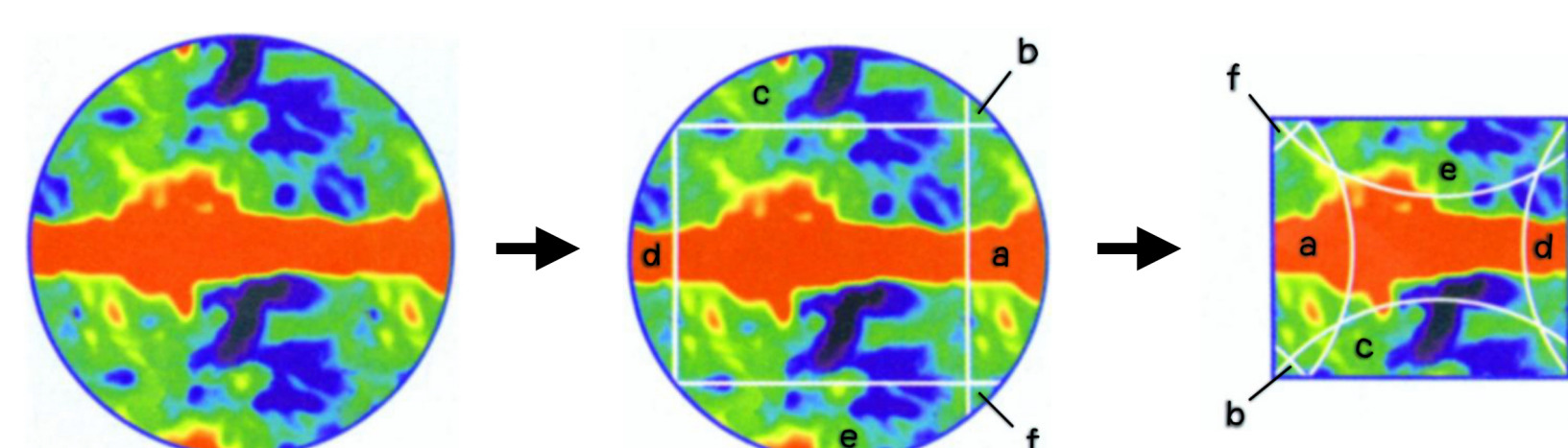


Fig: The visual circle can be divided into a fundamental domain and six other regions (a-f). Each of those regions corresponds to an area within the fundamental domain.

* Planck 2015 [3] disfavors small-scale topologies, but COMPACT [4] shows many possibilities remain unexplored.

- Galaxy clusters ”cosmic crystallography” (3-dim):

detecting multiple images of the same object, causing sharp peaks in a histogram of pair separations (non-normalized two-point correlation function), indicating a periodic structure in a multiply-connected Universe.

2 Literature Review [5]

Methods

- *Normalized temperature fluctuations* in CMB defined on the sphere $\mathbb{S}^2$:

$$f(\hat{n}) := (\delta T(\hat{n}) - \mu) / \sigma_0$$

- *Excursion set*:

$$\mathbb{E}(\nu) = \{\hat{n} \in \mathbb{S}^2 \mid f(\hat{n}) \geq \nu\}$$

- $\hat{n}$: direction of the sky

- μ : mean of δT

- σ_0 : standard deviation of δT

- ν : threshold

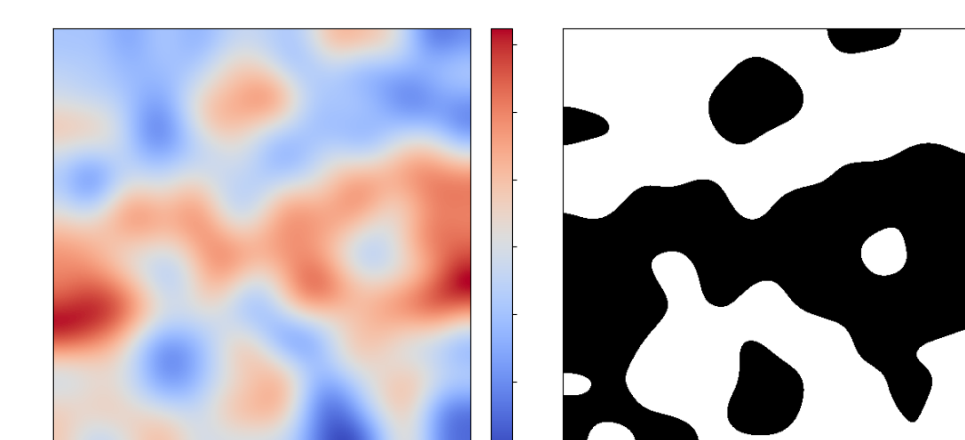


Fig: An example for an isotropic random field

- *Betti functionals*:

$$\beta_0(\nu) := \# \text{ components of } \mathbb{E}(\nu)$$

$$\beta_1(\nu) := \# \text{ topological holes of } \mathbb{E}(\nu)$$

* with $\nu \rightarrow -\nu$ ($\delta T \rightarrow -\delta T$) property: $\beta_0(\nu) = \beta_1(-\nu)$

* Betti numbers were originally used to distinguish topological spaces, e.g., for a 3-ball, $\beta_0 = 1, \beta_1 = 0$; for a 3-torus, $\beta_0 = 1, \beta_1 = 2$.

- *Normalized standard deviation of the CMB gradient field*:

$$\rho := \frac{\sigma_1}{\sigma_0} = \sqrt{\frac{2\pi e}{\beta_1(-1) - \beta_1(1)}}$$

$$\langle \rho(L) \rangle \approx 46.122 - 3.290L$$

can detect the cubic 3-torus size $L = c/H_0$, if the CMB is assumed to be homogeneous, isotropic and Gaussian.

Results

A comparison of the cubic 3-torus topology with the Planck CMB maps.

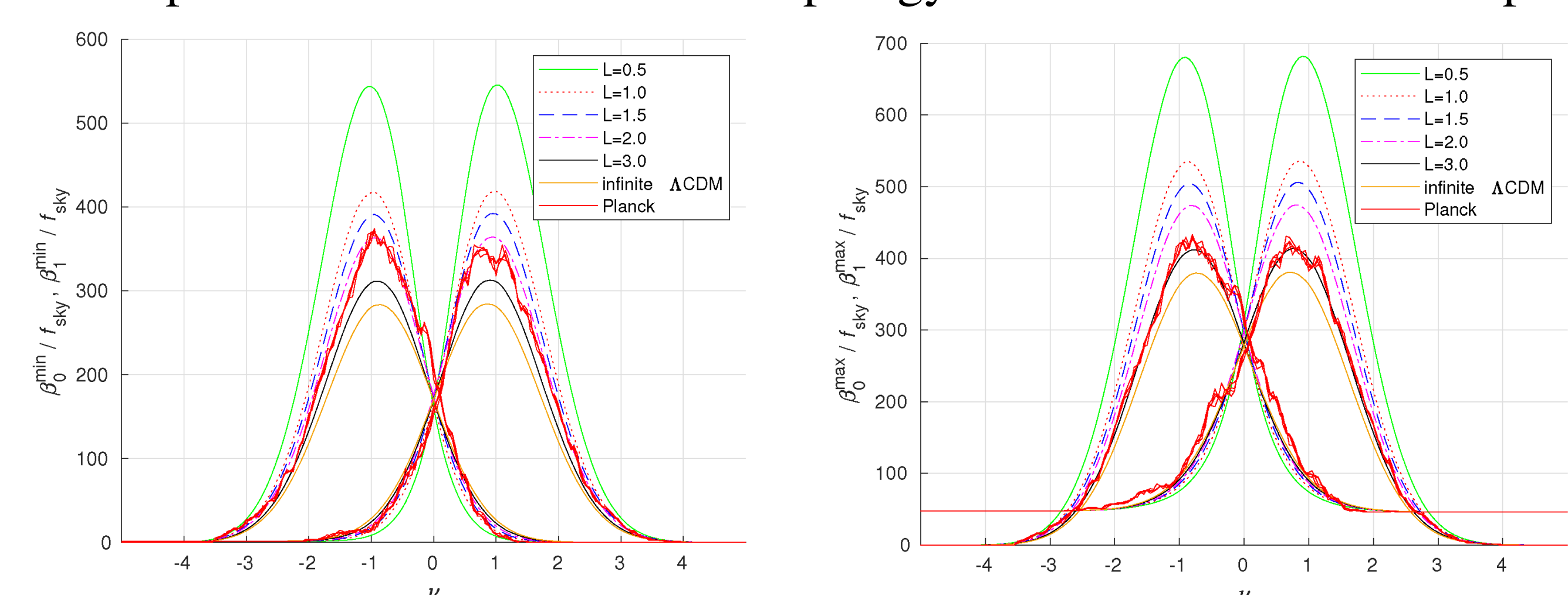


Fig: Plots for the 3-torus simulations, the infinite CDM model and the four Planck maps which are all subjected to the same mask. ($f_{\text{sky}} = \text{Area of the unmasked sky} / \text{Area of full sky}$)

The comparison shows that the curves derived from the Planck maps lie between the 3-torus models with side-length $L = 2.0$ and $L = 3.0$, which gives a further hint that our Universe has a possible *non-trivial* topology.

3 Discussion and Conclusions

- Betti numbers introduce a new way to detect topology, but inconclusive.
- CMB and galaxy anomalies may reflect errors or variance, not topology.
- Inconsistent models and biased surveys obscure topological detection.
- Complex topology may reveals the Universe’s true nature.

References

- [1] Jean-Pierre Luminet and Boudewijn F. Roukema. Topology of the universe: Theory and observations, 1999.
- [2] M. Lachieze-Rey and J. P. Luminet. Cosmic Topology. *Physics Reports*, 254(3):135–214, March 1995.
- [3] Planck Collaboration. Planck 2015 results. XVIII. Background geometry & topology. *Astronomy & Astrophysics*, 594:A18, October 2016.
- [4] COMPACT Collaboration. Promise of Future Searches for Cosmic Topology. *Phys. Rev. Lett.*, 132(17):171501, April 2024.
- [5] Ralf Aurich and Frank Steiner. Betti functionals as probes for cosmic topology. *Universe*, 10(5), 2024.